

Numerical Study of Ionization Conjecture in Density Functional Theory

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Abstract

Kohn-Sham functional theory is widely used in quantum chemistry and materials science. It is a powerful technique for representing atoms and molecules. The objective of this paper is to numerically investigate the maximum number of electrons that a charge can hold. For that purpose, in this study, we discretize Kohn-Sham equations **with PW exchange correlation, which falls within the scope of local spin density approximation**. We then implement the discretized model for the s and p-blocks of the periodic table in a Julia code. The execution revealed that very few charges could accommodate one additional electron. Since this model permits fractional occupation, it is clear that the majority of the charges strictly retain less than one extra electron. This showed that the aforementioned model is not suitable for negative charge representation.

1 Introduction

Plutôt commencer par la conjecture de l'ionsiation.

Petite intro Kohn-Sham, -> Kohn Sham étendue aussi ! Méthode des éléments finies -> Susi

The Kohn-Sham density functional theory is considered the backbone of the electronic structure theory [Lin2019]. This approach was first introduced by Kohn and Sham in the mid-1960s and was awarded the Nobel Prize in 1998. The idea behind this theory stems from transforming Schrödinger's equation into a system of non-interacting electron equations, which are based on an effective potential. This potential is a function of the electron density and is calculated in a self-consistent manner [Hohenberg1964; Kohn1965]. This theory is applied in a variety of fields, including quantum chemistry, condensed matter physics, and materials science.

The effective potential in the Kohn-Sham model contains what is known as the exchange-correlation functional of the electronic density. This function represents the electron-electron interaction, which is not fully included in the Hartree potential. It plays a crucial role in predicting the behavior of electrons, but there is no unique expression for this function so far, and it is still an active area of research. There are several types of exchange correlation potentials, including the Local Density Approximation (LDA), the Generalized Gradient Approximation (GGA), Hybrid functionals, Meta-GGA, etc. The choice of the type of exchange-correlation potential depends on the structure under study and the desired accuracy while taking into account the computational expenses [Perdew2001]. In this paper, one of the most commonly used approximations for correlation potential, Perdew and Wang's (PW) [Perdew1992], is considered. In fact, this

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last one belongs to the local spin density approximation (LSDA) [Kohn1965; Perdew1981].

Numerous attempts can be found in the literature to solve numerically the Kohn-Sham equations [Cances2018; Kauczor2011]. Recently, machine learning algorithms were also incorporated in solving these equations [Fiedler2022]. Numerically, the effective potential is calculated based on the obtained electronic density from the previous iteration, and this is repeated until convergence is achieved. Upon convergence, various properties of materials or molecules can be deduced, such as energy levels, total energy, and band structure. This gives materials science researchers a good prediction for the properties of the structure under study before it is tested experimentally.

The ionization theory deals with ions, where atoms or molecules gain or lose electrons. This theory is crucial in many disciplines to explain phenomena including the formation of plasma, chemical processes, and cosmic ray behavior. To our knowledge, for the Kohn-Sham model, there is no theoretical proof for the existence of ground-state electron density for negatively ionized ions. Thus, this study aims to numerically discover the efficiency of the Kohn-Sham equations in modeling negative ions.

The paper is structured as follows: Section 2 presents the extended the Kohn-Sham model with the associated Euler-Lagrange equations. Section ?? summarizes ionization conjecture results for various models representing atoms and molecules. Then, Section 4 details the derivation of the discretized self-consistent Kohn-Sham model, together with the algorithms used for implementing it. The obtained numerical results are stated and analyzed in Section ?. Lastly, Section ?? concludes the result.

2 Problem Formulation

2.1 Minimization Problem We consider a system of $N \in \mathbb{N}^*$ electrons interacting through Coulomb repulsion and attracted by a fixed nucleus of charge $Z \geq 0$ localized at the origin. Within the Born-Oppenheimer approximation of non-relativistic atoms [BO27], this system is described by the many-body electronic Hamiltonian

$$H_N := -\frac{1}{2} \sum_{i=1}^N \Delta_i - \sum_{i=1}^N \frac{Z}{|x_i|} + \sum_{1 \leq i < j \leq N} \frac{1}{|x_i - x_j|}. \quad (2.1)$$

As explained in the introduction, a central object of interest is the groundstate energy of H_N , obtained by minimizing the quadratic form $\langle \Psi | H_N \Psi \rangle$ over all the normalized fermionic states $\Psi \in \bigwedge_{i=1}^N H^1(\mathbb{R}^3, \mathbb{C}^2)$ and for the scalar product of $L^2(\mathbb{R}^3, \mathbb{C}^2)^N$.

Since a direct treatment of this variational problem is computationally intractable for large N , the extended Kohn-Sham density-functional theory (see for example [DG90]) replaces it with an auxiliary non-interacting system reproducing the exact electronic density. In this framework, $N > 0$ can be a real number, and the minimization problem reads

$$\underset{\gamma \in \mathcal{K}_N}{\text{minimize}} E_{Z,N}(\gamma) \quad (2.2)$$

where the energy functional is defined by

$$E_{Z,N}(\gamma) := \text{Tr} \left(-\frac{1}{2} \Delta \gamma \right) - Z \int_{\mathbb{R}^3} \frac{\rho_\gamma}{|\cdot|} + D(\rho_\gamma, \rho_\gamma) + E_{xc}(\gamma), \quad (2.3)$$

over the set of one-body density matrix

$$\mathcal{K}_N := \{ \gamma \in \mathcal{S}(L^2(\mathbb{R}^3, \mathbb{C}^2)) \mid \text{Tr}(\gamma) = N, \text{Tr}(-\Delta \gamma) < \infty, 0 \leq \gamma \leq 1 \}, \quad (2.4)$$

where $\mathcal{S}(L^2(\mathbb{R}^3, \mathbb{C}^2))$ is the set of self-adjoint operators on $L^2(\mathbb{R}^3, \mathbb{C}^2)$. The condition $0 \leq \gamma \leq 1$ means that all the eigenvalue of γ are between 0 and 1, and $\text{Tr}(-\Delta \gamma) := \text{Tr}(|\nabla| \gamma |\nabla|)$ with $|\nabla| := (-\Delta)^{\frac{1}{2}}$. In particular, γ is trace-class and

positive. Therefore, it is possible to define its kernel $\gamma(x, y) \in L^2(\mathbb{R}^3 \times \mathbb{R}^3, \mathbb{C}^{2 \times 2})$ defined through

$$\gamma f(x) = \int_{\mathbb{R}^3} \gamma(x, y) f(y) dy \quad (2.5)$$

for $f \in L^2(\mathbb{R}^3, \mathbb{C}^2)$ and $x \in \mathbb{R}^3$. We can define then the one-body reduced density $\rho_\gamma(x) := \text{Tr}_{\mathbb{C}^2} \gamma(x, x) \in L^1(\mathbb{R}^3)$. Note that, thanks to the Hoffman-Ostenhof inequality [HHA78]

$$\int_{\mathbb{R}^3} |\nabla \sqrt{\rho_\gamma}|^2 \leq \text{Tr}(-\Delta \gamma) < +\infty, \quad (2.6)$$

we have $\sqrt{\rho_\gamma} \in H^1(\mathbb{R}^3)$ and therefore by classical Sobolev embedding theorems, $\rho_\gamma \in L^3(\mathbb{R}^3)$.

In equation (2.3), the first term corresponds to the kinetic energy and is clearly well-defined for $\gamma \in \mathcal{K}_N$. The second term corresponds to the nucleus-electron attraction and is well-defined because $\rho_\gamma \in L^3(\mathbb{R}^3)$:

$$\int_{\mathbb{R}^3} \frac{\rho_\gamma(x)}{|x|} = \int_{|x| \geq 1} \frac{\rho_\gamma(x)}{|x|} + \int_{|x| < 1} \frac{\rho_\gamma(x)}{|x|} \quad (2.7)$$

$$\leq N + \|\rho_\gamma\|_{L^3} \left(\frac{8\pi}{3} \right)^{\frac{2}{3}} < +\infty. \quad (2.8)$$

The third term is the Coulomb energy defined by

$$D(\rho_\gamma, \rho_\gamma) := \frac{1}{2} \int_{\mathbb{R}^3} V^H(\rho_\gamma) \rho_\gamma \quad (2.9)$$

where $V^H(\rho) := \rho * \frac{1}{|\cdot|}$ is the Hartree potential. This is well defined because by interpolation $\rho_\gamma \in L^{6/5}(\mathbb{R}^3)$ and the Hardy-Littlewood-Sobolev inequality (see for example [Lie83]):

$$\int_{\mathbb{R}^3 \times \mathbb{R}^3} \frac{f(x)g(y)}{|x-y|} dx dy \leq C \|f\|_{L^{\frac{6}{5}}} \|g\|_{L^{\frac{6}{5}}}. \quad (2.10)$$

Note also, that the Hartree potential is the solution of the Poisson's equation

$$-\Delta V_H(\rho_\gamma) = 4\pi \rho_\gamma \geq 0. \quad (2.11)$$

Finally, the fourth term $E_{\text{xc}}(\gamma)$ is the exchange-correlation energy on which we come back later.

Because $\mathcal{S}(L^2(\mathbb{R}^3, \mathbb{C}^2)) \simeq \mathcal{S}(L^2(\mathbb{R}^3))^2$, any one-body density matrix $\gamma \in \mathcal{K}_N$ can be written as

$$\gamma = \begin{pmatrix} \gamma^{\uparrow\uparrow} & \gamma^{\uparrow\downarrow} \\ \gamma^{\downarrow\uparrow} & \gamma^{\downarrow\downarrow} \end{pmatrix}$$

where each $\gamma^{\sigma_1 \sigma_2}$ is a self-adjoint operator on $L^2(\mathbb{R}^3, \mathbb{C})$. In this work, we adopt the following assumption.

Assumption 2.1 (Collinear spin approximation). *We restrict the minimisation problem to the subset $\mathcal{K}_N^{\text{col}} := \{\gamma \in \mathcal{K}_N \mid \gamma^{\uparrow\downarrow} = \gamma^{\downarrow\uparrow} = 0\}$. The new problem is therefore*

$$\underset{\gamma \in \mathcal{K}_N^{\text{col}}}{\text{minimize}} E_{Z,N}(\gamma). \quad (2.12)$$

This approximation amounts to assuming that the spin-quantization axis is fixed and that the system does not exhibit any non-collinear spin behaviour. It is fully justified in the absence of spin-orbit coupling and external magnetic fields, in which case the Hamiltonian is invariant under spin rotations around a fixed axis, and the groundstate can always be chosen to be collinear.

Let us recall that the set $\mathcal{K}_N^{\text{col}}$ is exactly the set of $\gamma \in \mathcal{S}(L^2(\mathbb{R}^3, \mathbb{C}^2))$ such that it exists a set of orbitals (ϕ_i^σ) of $H^1(\mathbb{R}^3, \mathbb{C})$, satisfying the orthonormality condition $\int_{\mathbb{R}^3} \phi_i^{\sigma,*} \phi_j^\sigma = \delta_{ij}$, and a sequence (n_i^σ) of real numbers of $[0, 1]$, called the occupation numbers, such that

$$\gamma = \sum_{i \geq 1} \sum_{\sigma \in \{\uparrow, \downarrow\}} n_i^\sigma |\phi_i^\sigma\rangle \langle \phi_i^\sigma|. \quad (2.13)$$

Let us recall that in the usual, i.e the not extended version, Kohn-Sham, the minimization is done by replacing the condition $0 \leq \gamma \leq 1$ by the more restrictive condition $\gamma^2 = \gamma$, i.e γ is a projector. As a consequence, the occupation number are either 0 or 1, i.e an orbital is completely filled or completely unfilled and a one-body density matrix is equivalent to the data of the Slater determinants $\Phi = \sqrt{N!} \bigwedge_{i=1}^N \phi_i$. The advantage of the extended-Kohn Sham formulation is that it enables to consider a non-interger number of electrons which is well suited to track the maximal ionization number, and because the minimization is performed over a convex set. In the collinear setting, it is also possible to define the spin-reduced one-body density

$$\rho_\gamma^\sigma(x) = \gamma^{\sigma\sigma}(x, x) \quad (2.14)$$

such that $\rho_\gamma = \rho_\gamma^\uparrow + \rho_\gamma^\downarrow$

Let us now discuss about the exchange-correlation term $E_{xc}(\gamma)$. A complete review on exchange-correlation approximations can be found in [Tou22].

2.2 Euler-Lagrange equations From a variational viewpoint, solving the Kohn-Sham problem can be done in two equivalent ways. One may either discretize the energy functional and search for its minimize by setting the finite-dimentional gradient orthogonal to the tangent space of $\mathcal{K}_N$, or derive the Euler-Lagrange equations of the continuous problem and then discretize them. While both approaches lead to the same discrete Kohn-Sham equations in principle, the latter makes the underlying mathematical structure of the Kohn-Sham equations more transparent. Let us recall that the Euler-Lagrange equations to the problem 2.12 are

$$\left\{ \begin{array}{l} \text{Find } (\phi_i^\sigma, n_i^\sigma, \epsilon_i^\sigma) \in (H^1(\mathbb{R}^3) \times [0, 1] \times \mathbb{R})^N \text{ and } \epsilon_{Z,N,F} \leq 0 \text{ s.t.} \\ \int_{\mathbb{R}^3} \phi_i^\sigma \phi_j^\sigma = \delta_{ij}, \quad \sum_{i \geq 0} \sum_{\sigma \in \{\uparrow, \downarrow\}} n_i^\sigma = N, \\ \rho = \sum_{i \geq 0} \sum_{\sigma \in \{\uparrow, \downarrow\}} n_i^\sigma |\phi_i^\sigma|^2, \\ H_Z^\sigma(\rho) \phi_i^\sigma = \epsilon_i^\sigma \phi_i^\sigma, \\ n_i^\sigma = 1 \text{ if } \epsilon_i^\sigma < \epsilon_{Z,N,F}, \\ n_i^\sigma = 0 \text{ if } \epsilon_i^\sigma > \epsilon_{Z,N,F}, \\ n_i^\sigma \in [0, 1] \text{ if } \epsilon_i^\sigma = \epsilon_{Z,N,F}, \end{array} \right. \quad (2.15)$$

where $\epsilon_{Z,N,F}$ is called the Fermi level and can be seen as the Lagrange multiplier of the constraints $\text{Tr } \gamma = N$. The way the occupation number are chosen is nothing more than the usual *Aufbau* principle. $H_Z^\sigma(\rho)$ is the *mean-field hamiltonian* in the section σ and depends on the one-body reduced density. It is defined by

$$H_Z^\sigma(\rho) := -\frac{1}{2}\Delta - \frac{Z}{|\cdot|} + V^H(\rho) + v_{xc}^\sigma(\rho).$$

with the exchange-correlation potential

$$v_{xc}^\sigma(\rho) = \frac{d\varepsilon_{xc}^\sigma}{d\rho^\sigma}(\rho). \quad (2.16)$$

When no spin-polarization effect is taken into account as for LDA models, we simply note $v_{xc}(\rho) = v_{xc}^\uparrow(\rho) = v_{xc}^\downarrow(\rho)$.

Find a minimum to the problem 2.12 is therefore equivalent to solve the non linear eigenvalue problem 2.15. In section 3, we will recall what it is known about the existence of solution to these Euler-Lagrange equations.

2.3 Spherical symmetry In absence of external fields, the many-electron Hamiltonian is rotationally invariant. As a consequence, the ground state density of an atom is known to be spherically symmetric. It is therefore natural to restrict the variational problem to radial densities in the atomic case, and

later investigate what happens when this symmetry is broken, in particular for negatively charged ions for which no such symmetry result is theoretically guaranteed. We will refer to the following assumption when it is used

Assumption 2.2 (Spherical Symmetry). *The density corresponding to the minimizer γ of 2.12 depends only on the radial coordinate, i.e. $\rho_\gamma(x) = \rho_\gamma(|x|)$.*

Under this assumption, the Euler-Lagrange equations 2.15 decouple into independent radial problems. Let us recall how. The operator Δ can be decomposed in spherical coordinates

$$\Delta = \frac{1}{r} \frac{\partial^2}{\partial r^2} r + \frac{1}{r^2} \Delta_{\mathbb{S}^2}$$

where $\Delta_{\mathbb{S}^2}$ is the Laplace-Beltrami operator on $L_{\mathbb{S}}^2 := L^2(\mathbb{S}^2, d\omega)$, where $\mathbb{S}$ is the unit sphere of $\mathbb{R}^3$ and $d\omega := \sin(\theta)d\theta d\varphi$ is the Lebesgue measure on $\mathbb{S}^2$. It satisfies

$$\Delta_{\mathbb{S}^2} \mathcal{Y}_l^m = -l(l+1) \mathcal{Y}_l^m \quad (2.17)$$

where $\{\mathcal{Y}_l^m \mid l \geq 0, -l \leq m \leq l\}$ are the real spherical harmonics which forms an orthonormal basis of $L_{\mathbb{S}}^2$. Let us note for the following $\mathcal{H}^{s, \text{rad}} := H^s(\mathbb{R}^+, r^2 dr)$ for any $s \geq 0$. Using the canonical isomorphism

$$L^2(\mathbb{R}^3) \simeq \mathcal{H}^{0, \text{rad}} \otimes L_{\mathbb{S}}^2,$$

we obtain the standard decomposition

$$L^2(\mathbb{R}^3) = \bigoplus_{l \geq 0} \mathcal{H}_l^0, \quad \mathcal{H}_l^0 := \mathcal{H}^{0, \text{rad}} \otimes \text{Span}_{\mathbb{R}}((\mathcal{Y}_l^m)_{-l \leq m \leq l}) \quad (2.18)$$

in which the laplacian can be decomposed as

$$\Delta = \bigoplus_{l \geq 0} \Delta_l \otimes \text{Id}_{L_{\mathbb{S}}^2}, \quad \Delta_l := \frac{1}{r} \frac{\partial^2}{\partial r^2} r - \frac{l(l+1)}{r^2}. \quad (2.19)$$

Since a radial potential commutes with the rotation representation of $SO(3)$, the mean-field Hamiltonian commutes with $\Delta_{\mathbb{S}}$. As a consequence, one has the decomposition :

$$H_Z^\sigma(\rho) = \bigoplus_{l \geq 0} H_Z^{\sigma, l}(\rho), \quad H_{Z, N}^{\sigma, l}(\rho) := H_Z^\sigma(\rho)|_{\mathcal{H}_l^2}. \quad (2.20)$$

Using the standard isometry

$$u \in \mathcal{H}^{1, \text{rad}} \mapsto (r \in \mathbb{R}^+ \mapsto ru(r)) \in H_0^1(\mathbb{R}^+), \quad (2.21)$$

the previous problem 2.15 is equivalent to the following problem in $H_0^1(\mathbb{R}^+)$:

$$\left\{ \begin{array}{l} \text{Find } (u_{nl}^\sigma, n_{nl}^\sigma, \epsilon_{nl}^\sigma) \in (H_0^1(\mathbb{R}^+) \times [0, 1] \times \mathbb{R})^{\mathbb{N}} \text{ and } \epsilon_{Z, N, F} \leq 0 \text{ s.t.} \\ \int_{\mathbb{R}^+} u_{n_1 l}^\sigma u_{n_2 l}^\sigma = \delta_{n_1 n_2}, \quad \sum_{\sigma \in \{\uparrow, \downarrow\}} \sum_{n > l \geq 0} n_{nl}^\sigma = N, \\ \rho \text{ is given by equation (2.29)} \\ h_Z^{l, \sigma}(\rho) u_{nl}^\sigma(r) = \epsilon_{nl}^\sigma u_{nl}^\sigma(r), \\ n_{nl}^{\sigma \uparrow} = 1 \text{ if } \epsilon_{nl}^\sigma < \epsilon_{Z, N, F}, \\ n_{nl}^{\sigma \downarrow} = 0 \text{ if } \epsilon_{nl}^\sigma > \epsilon_{Z, N, F}, \\ n_{nl}^{\sigma \uparrow} \in [0, 1] \text{ if } \epsilon_{nl}^\sigma = \epsilon_{Z, N, F}, \end{array} \right. \quad (2.22)$$

where $h_Z^{l, \sigma}(\rho)$ is the H_0^1 -mean field hamiltonian, defined by

$$h_Z^{l, \sigma}(\rho) := -\frac{d^2}{dr^2} + \frac{l(l+1)}{2r^2} - \frac{Z}{r} + v_{\text{xc}}(\rho), \quad (2.23)$$

which is an operator on $L^2(\mathbb{R}^+, dr)$. For a fix $l \geq 0$, each solution u_{nl} of eq. 2.22 is indexed by the integer n starting from $l+1$, so as the occupation numbers

(n_{nl}) and the orbitals energy (ϵ_{nl}) . More details on this hamiltonian and the existence of (ϵ_{nl}) is given in the next

Therefore, all the computations in $\mathbb{R}^3$ are reduced to computations in the radial part. Indeed, the orbitals in $\mathbb{R}^3$ are

$$\phi_{nlm}^\sigma(r, \theta, \varphi) = \frac{u_{nl}^\sigma(r)}{r} g_{lm}^\sigma(\theta, \varphi)$$

for $n > l \geq 0$ and $-l \leq m \leq l$, where $(g_{ml})_{-l \leq m \leq l}$ are any orthonormal basis of $\text{Span}_{\mathbb{R}}((\mathcal{Y}_l^m)_{-l \leq m \leq l})$. The one-body density matrix is

$$\gamma = \sum_{\sigma \in \{\uparrow, \downarrow\}} \sum_{n > l \geq 0} \sum_{-l \leq m \leq l} n_{nl} |\phi_{nlm}^\sigma\rangle \langle \phi_{nlm}^\sigma| \quad (2.24)$$

$$= \sum_{\sigma \in \{\uparrow, \downarrow\}} \sum_{n > l \geq 0} n_{nl} \left| \frac{u_{nl}^\sigma(r)}{r} \right\rangle \left\langle \frac{u_{nl}^\sigma(r)}{r} \right| \otimes \sum_{-l \leq m \leq l} |g_{lm}^\sigma\rangle \langle g_{lm}^\sigma| \quad (2.25)$$

$$= \bigoplus_{\sigma \in \{\uparrow, \downarrow\}} \bigoplus_{l \geq 0} \left(\gamma_l^{\sigma, \text{rad}} \otimes \text{Id}_{\mathcal{H}_l} \right) \quad (2.26)$$

where the radial components of γ are operators on $\mathcal{H}^{\text{rad}}$ and defined by

$$\gamma_l^{\sigma, \text{rad}} := \sum_{n > l} n_{nl} \left| \frac{u_{nl}^\sigma(r)}{r} \right\rangle \left\langle \frac{u_{nl}^\sigma(r)}{r} \right|. \quad (2.27)$$

Then, the one-body density can be written

$$\rho^\sigma(r) = \frac{1}{4\pi} \sum_{l \geq 0} (2l+1) \gamma_l^{\sigma, \text{rad}}(r, r) \quad (2.28)$$

$$= \frac{1}{4\pi} \sum_{n > l \geq 0} (2l+1) n_{nl} \frac{|u_{nl}^\sigma(r)|^2}{r^2}. \quad (2.29)$$

3 Theoretical results

In this section, we first review the current theoretical results on the existence and uniqueness of minimizers of equation (2.12) for neutral atoms ($N = Z$) and positive ions ($N < Z$) in section 3.2. We then discuss the ionization conjecture for negative ions ($N > Z$) in section 3.3.

3.1 Mathematical insight of the H_0^1 -mean field hamiltonian First, let us recall the following basic fact on the H_0^1 -mean field hamiltonian $h_Z^{\sigma, l}(\rho)$ (see [Lew24a, Corollary 3.20] and [Lew24b, Theorem 5.40]) given for a certain class of potential V :

Theorem 3.1. *Let $V \in L^\infty(\mathbb{R}^+)$ and $l \in \mathbb{N}$. The radial operator*

$$h_{Z,V}^l := -\frac{1}{2} \frac{d^2}{dr^2} + \frac{l(l+1)}{2r^2} - \frac{Z}{r} + V. \quad (3.1)$$

is an unbounded operator on $L^2(\mathbb{R}^+)$ with domain $D(h_{Z,V}^l) = H^2(\mathbb{R}^+) \cap H_0^1(\mathbb{R}^+)$. Its essential spectrum is $\sigma_{\text{ess}}(h_{Z,V}^l) = [0, +\infty)$. Moreover, the sequence

$$\epsilon_{nl}(V) := \inf_{\substack{W \subset D(h_{Z,V}^l) \\ \dim W = n-l}} \max_{\substack{u \in W \\ \|u\|=1}} \langle u | h_{Z,V}^l u \rangle_{L^2(\mathbb{R}^+)} \quad (3.2)$$

is well-defined for all $n > l$ and is equal to

1. *either the $n-l^{\text{th}}$ eigenvalue of h_V^l counted with multiplicity if the rank of $\mathbb{1}_{(-\infty, 0)}(h_{Z,V}^l)$ is equal or greater than $n-l$,*
2. *0 otherwise.*

Remark 3.2. This theorem applies for the reduced-Hartree fock model because $V_{\text{har}}(\rho) \in L^\infty(\mathbb{R}^3)$ when $\rho \in L^1(\mathbb{R}^3) \cap L^3(\mathbb{R}^3)$ with the same inequality as in equation (2.8). For LDA models, one can show similar results but by extending the previous theorem to a wider class of potential V . For more complex DFT models, as GGA, the situation is harder so that, up to our knowledge, any kind of such results has been proven.

The previous theorem says in particular that even if the layer (nl) do not exists, even unfilled, the quantity ϵ_{nl} still exists and is just 0. When it is 0, either there exists a normalized associated eigenvector u_{nl} , or it does not. The next theorem enables to partially explained the Madelung rules in the ordering of the energies of the orbitals :

Lemma 3.3 (Ordering of eigenvalue). *Let $V \in L^\infty(\mathbb{R}^+)$ such that $-(rV(r))'' > 0$ on $(0, +\infty)$. Then, for every $n > l \geq 0$, we have*

$$\epsilon_{nl}(V) < \epsilon_{n+1,l}(V) < \epsilon_{n+1,l+1}(V). \quad (3.3)$$

when $\epsilon_{nl}(V)$ and $\epsilon_{n+1,l}(V)$ are eigenvalues of h_V^l .

The second inequality was proven in [BGM84], with another proof in [AB88], and is called the *BGM inequality*. The first one do not rely on the hypothesis of the concavity of $r \mapsto rV(r)$ and can be easily proved as it follows.

Proof. Let $l \in \mathbb{N}$ and $E < 0$ be an eigenvalue of h_V^l associated to two eigenvector u and v . Let us consider the Wronskian of u and v :

$$W(r) = u(r)v'(r) - u'(r)v(r).$$

Indeed, $h_V^l u = E$ is a second-order ordinary-differential equation with continuous coefficients when E is fixed. We have $W'(r) = 0$ and $W(0) = 0$ so the Wronskian is identially 0 on $(0, +\infty)$. In particular, the eigenspace associated to the eigenvalue E has a dimension equal to 1. Therefore, $\epsilon_{nl} < \epsilon_{n+1,l}$ when ϵ_{nl} is an eigenvalue of h_V^l . $\square$

In particular, we have

$$\epsilon_{1s}(V) < \epsilon_{2s}(V) < \min(\epsilon_{2p}(V), \epsilon_{3s}(V)) \leq \max(\epsilon_{2p}(V), \epsilon_{3s}(V)) < \epsilon_{3p}(V).$$

In this section, we provide some asymptotic analysis results which is usefull to understand better how to define appropriate grid.

Theorem 3.4. *Let u_{nl} and ϵ_{nl} be solution of equation (2.22) in the section $\mathcal{H}_l$. Then, for any $0 < \epsilon < 1$, we have the asymptotically decay*

1. *If $\epsilon_{nl} < 0$,*

$$|u_{nl}(r)| \leq Cr^{-\frac{N-Z}{\sqrt{-2\epsilon_{nl}}}} e^{\sqrt{-2\epsilon_{nl}}r}. \quad (3.4)$$

2. *If $\epsilon_{nl} = 0$*

$$|u_{nl}(r)| \leq Ce^{-2\sqrt{2(N-Z)}\sqrt{r}}. \quad (3.5)$$

Proposition 3.5.

$$\text{card} \{ (n', l') \in \mathbb{N}^2 \mid n' > l' \epsilon_{n'l'} = \epsilon_{nl} \} \leq \left\lfloor \frac{n-l}{2} \right\rfloor + l + 1 \quad (3.6)$$

3.2 Existence results for Extended Kohn-Sham models for $N \leq Z$

The study of the existence and uniqueness of ground states for the minimization problem equation (2.12) has been initiated in several works, but a complete mathematical understanding covering the full range of standard density functional approximations is still lacking. While rigorous results are available for reduced Hartree–Fock and local density approximations, the extension to more elaborate exchange–correlation functionals remains largely open.

In [Sol91], it has been proven that the reduced-Hartree Fock model have a groundstate, associated to an unique one-body reduced density :

Theorem 3.6 (Ground state for the rHF model). *Let $Z > 0$ and $N \leq Z$. Then the minimization problem equation (2.12) has a ground state $\gamma_{Z,N}^{0,\text{rHF}}$, and all the ground states share the same density $\rho_{Z,N}^{0,\text{rHF}} := \rho_{\gamma_{Z,N}^{0,\text{rHF}}}$. The mean-field Hamiltonian*

$$H_{Z,N}^{0,\text{rHF}} := -\frac{1}{2}\Delta - \frac{Z}{|\cdot|} + V^{\text{H}}(\rho_{Z,N}^{0,\text{rHF}}),$$

is a bounded-below self-adjoint operator on $L^2(\mathbb{R}^3, \mathbb{C})$ with an essential spectrum $\sigma_{\text{ess}}(H_{Z,N}^{0,\text{rHF}}) = \mathbb{R}_+$. Moreover, the ground state $\gamma_{Z,N}^{0,\text{rHF}}$ is of the form

$$\gamma_{Z,N}^{0,\text{rHF}} = 2 \mathbb{1}_{(-\infty, \epsilon_{Z,N,F}^{0,\text{rHF}})}(H_{Z,N}^{0,\text{rHF}}) + \delta_{Z,N}^{0,\text{rHF}},$$

where $\epsilon_{Z,N,F}^{0,\text{rHF}} \leq 0$ is the Fermi level, and $\delta_{Z,N}^{0,\text{rHF}}$ a self-adjoint operator on $L^2(\mathbb{R}^3, \mathbb{C})$ such that $0 \leq \delta_{Z,N}^{0,\text{rHF}} \leq 2$ and

$$\text{Ran}(\delta_{Z,N}^{0,\text{rHF}}) \subset \ker(H_{Z,N}^{0,\text{rHF}} - \epsilon_{Z,N,F}^{0,\text{rHF}}).$$

Finally, if $\epsilon_{Z,N,F}^{0,\text{rHF}}$ is negative and is not an accidentally degenerate eigenvalue of $H_{Z,N}^{0,\text{rHF}}$, then the ground state $\gamma_{Z,N}^{0,\text{rHF}}$ is unique.

A similar result has been given in [AC09] for LDA functionals. Let us recall that such functionals have the following form

$$E_{\text{xc}}^{\text{LDA}}(\rho) = \int_{\mathbb{R}^3} \epsilon_{\text{xc}}(\rho(r)) dr,$$

To show the existence of groundstate for extended Kohn-Sham models with LDA functionals, they do the following assumption.

Assumption 3.7. *We assume that the function ϵ_{xc} is a C^1 function from $\mathbb{R}_+$ into $\mathbb{R}_-$, twice differentiable on $\mathbb{R}_+^*$ and obeying the following conditions:*

- $\epsilon_{\text{xc}}(0) = 0$,
- $\epsilon'_{\text{xc}}(0) \leq 0$,
- $\exists 0 < \beta_- \leq \beta_+ < \frac{2}{3}$ s.t. $\sup_{\rho \in \mathbb{R}_+} \frac{|\epsilon'_{\text{xc}}(\rho)|}{\rho^{\beta_-} + \rho^{\beta_+}} < \infty$,
- $\exists 1 \leq \alpha < \frac{3}{2}$ s.t. $\limsup_{\rho \rightarrow 0^+} \frac{\epsilon_{\text{xc}}(\rho)}{\rho^\alpha} < 0$,
- $\exists \lambda > -1$ s.t. $\epsilon''_{\text{xc}}(\rho) \sim c\rho^\lambda$ as $\rho \rightarrow 0^+$.

Note that these assumptions are not restrictive because almost all LDA functionals satisfy them, so as the exact LDA functional. The theorem is

Theorem 3.8 (Ground state for the LDA models). *Let $Z > 0$ and $N \leq Z$. Let us consider a LDA model satisfying assumption 3.7. Then the minimization problem equation (2.12) has a ground state $\gamma_{Z,N}^{0,\text{LDA}}$, and all the ground states share the same density $\rho_{Z,N}^{0,\text{LDA}} := \rho_{\gamma_{Z,N}^{0,\text{LDA}}}$. The mean-field Hamiltonian*

$$H_{Z,N}^{0,\text{LDA}} := -\frac{1}{2}\Delta - \frac{Z}{|\cdot|} + V^{\text{H}}(\rho_{Z,N}^{0,\text{LDA}}) + v_{\text{xc}}(\rho_{Z,N}^{0,\text{LDA}}),$$

is a bounded-below self-adjoint operator on $L^2(\mathbb{R}^3, \mathbb{C})$ with an essential spectrum $\sigma_{\text{ess}}(H_{Z,N}^{0,\text{LDA}}) = \mathbb{R}_+$, and $v_{\text{xc}}(\rho_{Z,N}^{0,\text{LDA}})$ a bounded below self-adjoint operator on $L^2(\mathbb{R}^3, \mathbb{C}^2)$. Moreover, the ground state $\gamma_{Z,N}^{0,\text{LDA}}$ is of the form

$$\gamma_{Z,N}^{0,\text{LDA}} = 2 \mathbb{1}_{(-\infty, \epsilon_{Z,N,F}^{0,\text{LDA}})}(H_{Z,N}^{0,\text{LDA}}) + \delta_{Z,N}^{0,\text{LDA}},$$

where $\epsilon_{Z,N,F}^{0,\text{LDA}} \leq 0$ is the Fermi level, and $\delta_{Z,N}^{0,\text{LDA}}$ a self-adjoint operator on $L^2(\mathbb{R}^3, \mathbb{C})$ such that $0 \leq \delta_{Z,N}^{0,\text{LDA}} \leq 2$ and

$$\text{Ran}(\delta_{Z,N}^{0,\text{LDA}}) \subset \ker(H_{Z,N}^{0,\text{LDA}} - \epsilon_{Z,N,F}^{0,\text{LDA}}).$$

In [Gon14], this result has been extended to LSDA functionals. In [AC09], the authors also show a results for some GGA functionals, with unpolarized spins, but with strong assumptions and only for a single pair of electrons. Up to our knowledge, proving equivalent theorems for more complicated DFT functionals as meta-GGA or hybrids is still a challenge.

3.3 The Ionization Conjecture Let $Z \geq 0$ and denote by $N_c(Z)$ the maximal number of electrons that an atom with nuclear charge Z can bind. More precisely, $N_c(Z)$ is defined as the largest real number for which the many-body Hamiltonian $H_{N_c(Z)}$, introduced in equation (2.1), admits a ground state. Since $N_c(Z) \geq Z$, this quantity characterizes the possible existence of negative atomic ions, namely atoms carrying an excess negative charge.

The central question is then the following: *how many electrons can remain bound to a nucleus, and how many may escape to infinity?* From both a physical and a mathematical viewpoint, this question is fundamental, as it is intimately related to the existence of ground states and to the stability of many-electron systems. The *ionization conjecture* asserts that a neutral atom can bind only an uniformly bounded number of additional electrons.

Conjecture 3.9 (Ionization conjecture). *There exists a constant $C > 0$, independent of Z , such that*

$$N_c(Z) \leq Z + C.$$

This conjecture is strongly supported by experimental observations and is widely believed to hold with $C = 2$. Nevertheless, despite decades of research, no complete and fully rigorous proof is known within realistic quantum-mechanical models. Establishing such a bound remains one of the central open problems in the mathematical theory of atoms.

A first major step was to show that $N_c(Z)$ is finite, i.e. that a nucleus cannot bind an infinite number of excess electrons. This result was proved independently by Sigal [Sig82] and Ruskai [Rus82]. Subsequent works focused on obtaining quantitative upper bounds on $N_c(Z)$. The best-known *non-asymptotic* bounds are obtained by combining Lieb’s result from 1984 [Lie84b; Lie84a] with the improvement due to Nam [Nam12].

Theorem 3.10. *For all $Z \geq 0$,*

$$N_c(Z) < \begin{cases} 2Z + 1, & \text{if } Z \leq Z_{\text{Nam}}, \\ 1.22 Z + 3Z^{1/3}, & \text{if } Z \geq Z_{\text{Nam}}, \end{cases}$$

where $Z_{\text{Nam}} > 1$ is the unique solution of

$$1.22 Z_{\text{Nam}} + 3Z_{\text{Nam}}^{1/3} = 2Z_{\text{Nam}} + 1,$$

numerically given by $Z_{\text{Nam}} \approx 5.51$.

While these bounds are effective for all values of Z , they do not capture the asymptotic behavior of $N_c(Z)$ for large atomic numbers. In the limit $Z \rightarrow +\infty$, sharper estimates were obtained in 1990 through different methods by Fefferman and Seco [FS90], and independently by Seco, Sigal, and Solovej [SSS90].

Theorem 3.11. *As $Z \rightarrow +\infty$,*

$$N_c(Z) \leq Z + O(Z^{5/7}).$$

Beyond the full many-body Schrödinger model, the ionization conjecture also provides a stringent test for approximate theories of electronic structure, in particular those arising in density functional theory (DFT). A natural question is whether such models are able to describe negative ions.

For the simplest approximation, this question is completely understood. Lieb and Simon [LS77] proved that the Thomas–Fermi model admits a minimizer if and only if $N \leq Z$. As a consequence, this model is incapable of describing negative ions. A more realistic behavior is exhibited by the Thomas–Fermi–von

Weizsäcker (TFW) model. In this case, the energy functional admits a minimizer if and only if $N \leq N_c(Z)$ for some critical value satisfying

$$Z < N_c(Z) \leq Z + C,$$

as shown in [BBL81; BL85]. The strict lower bound $N_c(Z) > Z$ indicates that the TFW model can accommodate a limited excess negative charge.

Finally, the ionization conjecture 3.9 was proved by Solovej for the reduced Hartree–Fock model in 1991 [Sol91], and later for the full Hartree–Fock model in 2003 [Sol04]. For more elaborate Kohn–Sham models, such as the local density approximation (LDA) or generalized gradient approximations (GGA), the ionization conjecture remains open. A comprehensive review of the ionization problem in quantum mechanics can be found in [Nam12].

4 Discretization

In this section, we provide the numerical method used to solve the problem equation (2.12) under the assumption 2.2. First, we discuss about the variational approximation with an abstract basis in section 4.1, and gives its matrix formulation in section 4.2 and the associated discretized Euler-Lagrange equations in section 4.3. Then, we describe the basis which lays on the finite-element method in section 4.4. Finally, we described the SCF algorithm used in ??.

4.1 Variational Approximation To discretize problem 2.12, we use a variational approximation. Under assumption 2.2, this amounts to discretizing the radial one-body density matrix $\gamma_l^{\sigma, \text{rad}}$ for each angular momentum channel $l \geq 0$ and spin $\sigma \in \uparrow, \downarrow$. To this end, we introduce a linearly independent family $(\chi_i)_{1 \leq i \leq N_h}$ of $N_h \in \mathbb{N}^*$ functions belonging to $\mathcal{H}^{1, \text{rad}}$, and we define

$$\mathcal{X}_{N_h} := \text{Span}(\chi_i \mid 1 \leq i \leq N_h) \subseteq \mathcal{H}^{1, \text{rad}}.$$

This finite-dimensional space is used to discretize the orthonormal radial orbitals (u_{nl}^σ) . The choice of the basis functions (χ_i) must be made carefully in order to ensure a controlled approximation error between the exact orbitals, which solve equation (2.22), and their discrete counterparts, obtained as solutions of the discretized problem. Moreover, this error should converge to zero as the dimension N_h of the approximation space tends to infinity. In section 4.4, we describe the specific family (χ_i) that we employ and recall standard approximation results ensuring the convergence of this discretization scheme. As a consequence, we seek for radial orbitals in $\mathcal{X}_{N_h}$, i.e orbitals of the form

$$u_{nl}^\sigma(r) = \sum_{i=1}^{N_h} U_i^{nl\sigma} \chi_i(r), \quad (4.1)$$

where the coefficients $U_i^{nl\sigma} \in \mathbb{R}$ are chosen so as to enforce the orthonormality constraints on the family of orbitals (u_{nl}^σ) . Because the full orbitals in $\mathbb{R}^3$ are

$$\phi_{nlm}^\sigma(r, \theta, \varphi) = \frac{u_{nl}^\sigma(r)}{r} \mathcal{Y}_l^m(\theta, \varphi)$$

our variational methods falls in the scope of the so-called *numerical atomic orbitals* (NAOs). Accordingly, the minimization problem 2.12 is performed over the discrete subset $\mathcal{K}_N^{\text{dis}} \subset \mathcal{K}_N$ consisting of density matrices that can be written as

$$\gamma_l^{\sigma, \text{rad}} = \sum_{l=0}^{l_{Z,N}^{\max}} \sum_{n=l+1}^{n_{Z,N}^{\max}} n_{nl} \sum_{i,j=1}^{N_h} U_i^{nl\sigma} U_j^{nl\sigma} \left| \frac{\chi_i(r)}{r} \right\rangle \left\langle \frac{\chi_j(r)}{r} \right|. \quad (4.2)$$

Any $\gamma \in \mathcal{K}_N^{\text{dis}}$ therefore depends only on the pair (U, n) and can be written explicitly as $\gamma[U, n]$.

4.2 Matrix formulation Let us give now the matrix formulation of the problem 2.12 using the basis $(\chi_i)_{1 \leq i \leq N_h}$. Let $\gamma[U, n] \in \mathcal{K}_N^{\text{dis}}$.

Orthonormality conditions First, the orthonormality conditions of the radial orbitals (u_{nl}^σ) become conditions on the coefficients $U^{nl\sigma}$:

$$\delta_{n_1 n_2} = \int_0^{+\infty} u_{n_1 l}^\sigma u_{n_2 l}^\sigma = (U^{n_1 l \sigma})^T \mathbb{M}_0 U^{n_2 l \sigma}, \quad (4.3)$$

where the mass matrix, a.k.a. the overlap matrix, $\mathbb{M}_0 \in \mathbb{R}^{N_h \times N_h}$ is

$$\mathbb{M}_{0,ij} := \int_0^\infty \chi_i \chi_j. \quad (4.4)$$

One-body density The density becomes

$$\rho_\gamma^\sigma(r) = \frac{1}{4\pi r^2} \sum_{l=0}^{l_{Z,N}^{\max}} \sum_{n=l+1}^{n_{Z,N}^{\max}} (2l+1) n_{nl}^\sigma \sum_{i,j=1}^{N_h} U_i^{nl\sigma} U_j^{nl\sigma} \chi_i(r) \chi_j(r) \quad (4.5)$$

$$= \frac{1}{4\pi r^2} \sum_{i,j=1}^{N_h} \mathbb{D}_{ij}^\sigma \chi_i(r) \chi_j(r) \quad (4.6)$$

where the coefficients

$$\mathbb{D}_{ij}^\sigma := \sum_{l=0}^{l_{Z,N}^{\max}} (2l+1) \sum_{n=l+1}^{n_{Z,N}^{\max}} n_{nl}^\sigma U_i^{nl\sigma} U_j^{nl\sigma} \quad (4.7)$$

can be seen as the discretization coefficients of ρ_γ^σ in the basis $(\chi_i \otimes \chi_j)_{1 \leq i,j \leq N_h}$ of $H_0^1(\mathbb{R}^+) \otimes H_0^1(\mathbb{R}^+)$. Any quantity $f(\rho_\gamma)$ depending on ρ_γ can therefore be seen as a quantity depending on $\mathbb{D}$, i.e we note $f[\mathbb{D}] := f(\rho_\gamma)$.

Kinetic energy The kinetic energy becomes

$$E_{\text{kin}}(\gamma) := \text{Tr} \left(-\frac{1}{2} \Delta \gamma \right) = \sum_{\substack{\sigma \in \{\uparrow, \downarrow\} \\ 0 \leq l \leq l_{Z,N}^{\max}}} (2l+1) \text{Tr} \left(-\frac{1}{2} \Delta_l \gamma_l^{\sigma, \text{rad}} \right) \quad (4.8)$$

$$= \sum_{\substack{\sigma \in \{\uparrow, \downarrow\} \\ 0 \leq l \leq l_{Z,N}^{\max}}} \sum_{n=l+1}^{n_{Z,N}^{\max}} (2l+1) n_{nl}^\sigma (U^{nl\sigma})^T \mathbb{K} U^{nl\sigma} \quad (4.9)$$

where the kinetic matrix $\mathbb{K} \in \mathbb{R}^{N_h \times N_h}$ is the sum of the stiffness matrix $\mathbb{A} \in \mathbb{R}^{N_h \times N_h}$ and the weighted mass matrix $\mathbb{M}_{-2} \in \mathbb{R}^{N_h \times N_h}$

$$\mathbb{K} := \frac{1}{2} \mathbb{A} + \frac{1}{2} l(l+1) \mathbb{M}_{-2}, \quad (4.10)$$

which are defined by

$$\mathbb{A}_{ij} := \int_0^\infty \chi_i' \chi_j', \quad \mathbb{M}_{-2,ij} := \int_0^\infty \frac{\chi_i(r) \chi_j(r)}{r^2} dr. \quad (4.11)$$

Electron-Nucleous interaction Next, the electron-nucleous attraction energy is

$$E_{\text{ne}}(\rho_\gamma) := -Z \int_{\mathbb{R}^3} \frac{\rho_\gamma}{|\cdot|} = -Z \sum_{\sigma \in \{\uparrow, \downarrow\}} \sum_{i,j=1}^{N_h} \mathbb{D}_{ij}^\sigma \mathbb{M}_{-1,ij} \quad (4.12)$$

where the weighted mass matrix $\mathbb{M}_{-1} \in \mathbb{R}^{N_h \times N_h}$ is defined by

$$\mathbb{M}_{-1,ij} = \int_0^\infty \frac{\chi_i(r) \chi_j(r)}{r} dr. \quad (4.13)$$

Hartree potential $V^H(\rho_\gamma)$ The Hartree potential can be analytically written :

$$V^H(\rho_\gamma)(r) = 4\pi \left(\frac{1}{r} \int_0^r \rho_\gamma(s) s^2 ds + \int_r^{+\infty} \rho_\gamma(s) s ds \right). \quad (4.14)$$

It is solution of the partial differential equation

$$\begin{cases} \text{Find } V(\rho) \in ? \text{ s.t} \\ -\frac{1}{r} \frac{d^2}{dr^2} (rV(\rho)) = 4\pi \rho_\gamma, \\ \lim_{r \rightarrow 0^+} rV(\rho)(r) = 0, \\ \lim_{r \rightarrow +\infty} rV(\rho)(r) = \int_{\mathbb{R}^3} \rho_\gamma = N. \end{cases} \quad (4.15)$$

Let us consider a continuous function $\theta : [0, +\infty) \rightarrow \mathbb{R}$ such that $\theta(0) = 0$, $\theta' \in L^2$, and $\lim_{r \rightarrow +\infty} \theta(r) = 1$. We seek a solution of equation (4.15) with the following form $V^H(\rho) = \frac{N\theta}{r} + \frac{w(r)}{r}$ and $w \in H_0^1(\mathbb{R}^+)$. Plugging it in equation (4.15), we obtain

$$\begin{cases} \text{Find } w \in H_0^1(\mathbb{R}^+) \text{ s.t} \\ -\frac{d^2}{dr^2} w(r) = 4\pi r \rho_\gamma(r) + N \frac{d^2}{dr^2} \theta(r). \end{cases} \quad (4.16)$$

Using a variational method with the same basis $(\chi_i)_{1 \leq i \leq N_h}$, we obtain the following finite-dimentional linear problem

$$\begin{cases} \text{Find } \mathbb{W}[\mathbb{D}] \in \mathbb{R}^{N_h} \text{ s.t} \\ \mathbb{A} \mathbb{W}[\mathbb{D}] = \mathbb{B}[\mathbb{D}] - \Theta. \end{cases} \quad (4.17)$$

where $\Theta, \mathbb{B}[\mathbb{D}] \in \mathbb{R}^{N_h}$ are defined respectively by

$$\Theta_m := N \int_0^{+\infty} \chi'_m \theta', \quad (4.18)$$

and

$$\mathbb{B}_m[\mathbb{D}] := 4\pi \int_0^{+\infty} r \rho_\gamma(r) \chi_m(r) dr = \sum_{\sigma \in \{\uparrow, \downarrow\}} \sum_{i,j=1}^{N_h} \mathbb{D}_{ij}^\sigma \mathbb{F}_{ijm} \quad (4.19)$$

with $\mathbb{F} \in \mathbb{R}^{N_h \times N_h \times N_h}$ the tensor defined by

$$\mathbb{F}_{ijm} := \int_0^{+\infty} \frac{\chi_i(r) \chi_j(r) \chi_m(r)}{r} dr. \quad (4.20)$$

Once the coefficients $\mathbb{W}[\mathbb{D}]$ of the discretization of w found, one get the Hartree potential $V^H(\rho_\gamma)$ and then the Hartree energy :

$$D(\rho_\gamma, \rho_\gamma) = \int_{\mathbb{R}^3} V^H(\rho_\gamma) \rho_\gamma = \sum_{\substack{\sigma \in \{\uparrow, \downarrow\} \\ 1 \leq i,j \leq N_h}} \left[\sum_{m=1}^{N_h} \mathbb{D}_{ij}^\sigma \mathbb{F}_{ijm} \mathbb{W}[\mathbb{D}]_m + N \mathbb{D}_{ij}^\sigma \mathbb{M}_{\theta,ij} \right] \quad (4.21)$$

where $\mathbb{M}_\theta \in \mathbb{R}^{N_h \times N_h}$ is the weighted mass matrix

$$\mathbb{M}_{\theta,ij} := \int_0^{+\infty} \frac{\chi_i(r) \chi_j(r) \theta(r)}{r} dr. \quad (4.22)$$

Total energy Gathering all together, the total energy for an element $\gamma \in \mathcal{K}_N^{\text{dis}}$ is

$$E_{Z,N}(\rho_\gamma) = E_{Z,N}^{\text{dis}}[n, U] := \sum_{\substack{\sigma \in \{\uparrow, \downarrow\} \\ 0 \leq l \leq l_{Z,N}^{\text{max}} \\ l < n \leq n_{Z,N}^{\text{max}}}} (2l+1) n_{nl} (U^{nl\sigma})^T \times \\ \left[\frac{1}{2} \mathbb{A} + \frac{l(l+1)}{2} \mathbb{M}_{-2} - Z \mathbb{M}_{-1} + \mathbb{F} : \mathbb{W}[\mathbb{D}] + N \mathbb{M}_\theta \right] U^{nl\sigma}, \quad (4.23)$$

and the discretized problem is

$$\underset{(n,U) \in \mathcal{M}_N^{\text{Dis}}}{\text{minimize}} \ E_{Z,N}^{\text{dis}}[n,U], \quad (4.24)$$

where the minimization is performed over the set

$$\mathcal{M}_N^{\text{Dis}} := \left\{ (n,U) \in [0,1]^{n_{Z,N}^{\max} \times (l_{Z,N}^{\max}+1) \times 2} \times \mathbb{R}^{n_{Z,N}^{\max} \times (l_{Z,N}^{\max}+1) \times 2 \times N_h \text{ s.t.}} \right. \\ \left. \sum_{\substack{\sigma \in \{\uparrow, \downarrow\} \\ 0 \leq l \leq l_{Z,N}^{\max} \\ l < n \leq n_{Z,N}^{\max}}} n_{nl}^{\sigma} = N \quad \text{and} \quad (U^{n_1 l \sigma})^T \mathbb{M}_0 U^{n_2 l \sigma} = \delta_{n_1 n_2}} \right\} \quad (4.25)$$

4.3 Discretized Euler-Lagrange equations The corresponding discretized Euler-Lagrange equations are

$$\begin{cases} \text{Find } (n,U,\epsilon) \in (\mathcal{M}_N^{\text{Dis}} \times \mathbb{R}^{n_{Z,N}^{\max} \times (l_{Z,N}^{\max}+1) \times 2}) \text{ and } \epsilon_{Z,N,F} \leq 0 \text{ s.t.} \\ \nabla_{U^{nl\sigma}} E_{Z,N}^{\text{dis}}[n,U] = \epsilon_{nl}^{\sigma} \mathbb{M}_0 U^{nl\sigma}, \\ n_{nl}^{\sigma} = 1 \text{ if } \epsilon_{nl}^{\sigma} < \epsilon_{Z,N,F}, \\ n_{nl}^{\sigma} = 0 \text{ if } \epsilon_{nl}^{\sigma} > \epsilon_{Z,N,F}, \\ n_{nl}^{\sigma} \in [0,1] \text{ if } \epsilon_{nl}^{\sigma} = \epsilon_{Z,N,F}, \end{cases} \quad (4.26)$$

where the gradient $\nabla_{U^{nl\sigma}} E_{Z,N}^{\text{dis}}[n,U]$ is given by

$$\nabla_{U^{nl\sigma}} E_{Z,N}^{\text{dis}}[n,U] = \mathbb{H}_Z^{\text{dis}}[\mathbb{D}] U^{nl\sigma} \quad (4.27)$$

with $\mathbb{H}_Z^{\text{dis},\sigma,l}[\mathbb{D}]$ being the discretized radial mean-fiel for spin σ and section l

$$\mathbb{H}_Z^{\text{dis},\sigma,l}[\mathbb{D}] := \frac{1}{2} \mathbb{A} + \frac{l(l+1)}{2} \mathbb{M}_{-2} - Z \mathbb{M}_{-1} + \mathbb{F} : \mathbb{W}[\mathbb{D}] + N \mathbb{M}_{\theta}.$$

This is a non-linear generalized eigen value problem, where the nonlinearity arises from the vector $\mathbb{W}[\mathbb{D}]$, which is defined as the solution of equation (4.17).

4.4 Finite Element method In this subsection, the choice of the family $(\chi_i)_{1 \leq i \leq N_h}$ falls within the broad class of finite element methods (FEM). FEM relies on basis functions with compact support, typically restricted to one or two mesh intervals in one dimension for a given grid. This strong locality leads to sparse discretization matrices and allows for an efficient and scalable numerical implementation. Moreover, the use of compactly supported basis functions enables local mesh refinement, which is particularly well suited to capture sharp spatial features such as the nuclear cusp, while keeping a coarser resolution in smoother regions.

[Leh19b] [Leh19c] [Leh20] [Leh23b] [Leh23a] [Leh19a]

5 Algorithms

In this section, we discuss the different algorithms to solve the Euler-Lagrange equation (4.26).

5.1 Self-Consistent iterations To solve the non-linear generalized eigenvalue problem equation (4.26), we used an iterative scheme. The one used is the self-consistent algorithm ODA (optimal damping algorithm) introduced in [CL00]. Let us describe how it works.

At step $m \geq 0$, do :

1. Compute the Hartree matrix $V^H[\mathbb{D}^m]$, and, for each $\sigma \in \{\uparrow, \downarrow\}$, the matrix $V_{\text{xc}}^{\sigma}[\mathbb{D}^m]$ according to

- For each $\sigma \in \{\uparrow, \downarrow\}$ and section $0 \leq l \leq l_{\max}$, compute the full discretized mean-field hamiltonian

$$\mathbb{H}^{\text{dis}, \sigma, l}[\mathbb{D}^m] = \frac{1}{2}\mathbb{A} + \frac{l(l+1)}{2}\mathbb{M}_{-2} - Z\mathbb{M}_{-1} + \mathbb{F} : \mathbb{W}[\mathbb{D}] + N\mathbb{M}_\theta, \quad (5.1)$$

and then solved the generalized finite-dimensional eigenvalue problem

$$\begin{cases} \text{Find } (U^{l\sigma}, \epsilon_l^\sigma) \in \mathbb{R}^{N_h \times N_h} \times \mathbb{R}^{N_h} \text{ s.t.} \\ \mathbb{H}^{\text{dis}, \sigma, l}[\mathbb{D}^m]U^{nl\sigma} = \epsilon_{nl}^\sigma \mathbb{M}_0 U^{nl\sigma} \end{cases}. \quad (5.2)$$

- Compute the occupation numbers (n_{nl}^σ) according to the Aufbau principle described below.

Initialized $N_{\text{left}}^1 = N$ and $i = 0$. Repeat until it stops :

- Increase $i = i + 1$.
- Find all the indices, noted I_i corresponding to the i -th slowest eigenvalue counted without multiplicity.
- Compute

$$N_{\text{need}}^i = \sum_{(nl\sigma) \in I_i} (2l+1).$$

- i. If $N_{\text{need}}^i \geq N_{\text{left}}^i$, then fills

$$n_{nl}^\sigma = 1$$

for every $(nl\sigma) \in I_i$. Update $N_{\text{left}}^{i+1} = N_{\text{left}}^i - N_{\text{need}}^i$. If $N_{\text{left}}^{i+1} = 0$, stops.

- ii. Otherwise, if $I_i = 1$, then fill $n_{nl}^\sigma = \frac{N_{\text{left}}^i}{2l+1}$. If $I_i > 1$ Note that during this step, you compute the energies.

- If it has not been done during the previous steps, update the current energy :

$$E^{\text{kin}, m} = \quad (5.3)$$

These energies has always been computed in step 3 if a degeneracy has been encountered.

- Compute the matrix $\mathbb{D}^{n+1, *}$ of the trial density :

$$\mathbb{D}^{n+1, *} = \quad (5.4)$$

- Minimize the functional

$$t \in [0, 1] \mapsto E(t\gamma^n + (1-t)\gamma^*) \in \mathbb{R} \quad (5.5)$$

where γ^* is the density matrix associated to the density $\rho^{n+1, *}$. See the next section 5.2 for more details on the procedure of this step. We note t^n the minimizer. Then, compute the matrix density of the next-step density

$$\mathbb{D}^{\sigma, n+1} = t^n \mathbb{D}^{\sigma, n} + (1-t^n) \mathbb{D}^{n+1, *} \quad (5.6)$$

- Update the energy

- If $||\mathbb{D}_{n+1} - \mathbb{D}_n|| < \epsilon$

5.2 Details on the minimization during ODA In this subsection, we describes how to minimize

5.3 Shooting method When getting close to the ionization number $N_c(Z)$, the last bounded eigenvalue ϵ converges towards 0. As a consequence, the associated eigenvector is more and more delocalized creating some troubles in the previous methodology. Indeed, one has to pick a higher and higher $R_{\max}$ which for the same amount of points leads to a decrease in the precision of the inner bounded layers which remains close to 0, and some numerical issue in the resolution of the eigen problem. To get a better precision, we use the following strategy. First, we use the previous methods to get close to $N_c(Z)$. When adding some extra small amount electrons, the occupation number of the inner layer will remain the same, so that they are frozen.

6 Numerical Results

7 Conclusion

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A Asymptotic analysis

The goal of this section is to prove theorem 3.4. For this, we consider first the more general radial problem

$$\begin{cases} \text{Find } u \in H_0^1(\mathbb{R}^+) \\ -\frac{1}{2} \frac{d^2}{dr^2} u + (V - E)u = 0 \end{cases} \quad (\text{A.1})$$

where $V : \mathbb{R}^+ \rightarrow \mathbb{R}$ is a Kato potential such that $V(r) \xrightarrow{r \rightarrow +\infty} 0$, and $E < 0$ a real in the point spectrum of $H^V := -\frac{1}{2} \frac{d^2}{dr^2} + V$. Let us recall that we want to proof a bound of the form

$$|u(r)| \leq C e^{-f(r)} \quad (\text{A.2})$$

for $r \geq R$ with R big enough, $C > 0$ a constant and $f : [R, +\infty) \rightarrow \mathbb{R}$ a function. The optimal f is well known in the litterature and is called *the Agmon metric* defined as follow.

Definition A.1 (Agmon metric). *The Agmon metric for the problem A.1 is defined by*

$$\Phi_E^{R_0}(r) := \int_{R_0}^r \sqrt{2(V(s) - E)_+} ds, \quad (\text{A.3})$$

where $r \geq R_0 \geq 0$ and $(x)_+ = \max(x, 0)$ for any $x \in \mathbb{R}$.

Remark A.2. In the litterature, for the problem

$$\begin{cases} \text{Find } u \in H^1(\mathbb{R}^d) \\ -\frac{1}{2} \Delta u + (V - E)u = 0, \end{cases} \quad (\text{A.4})$$

the usual definition of the Agmon metric is

$$d_E(x, y) := \inf_{\substack{\gamma : [0, 1] \rightarrow \mathbb{R}^d \text{ A.C.} \\ \gamma(0) = x, \gamma(1) = y}} \int_0^1 \sqrt{2(V(\gamma(t)) - E)_+} |\gamma'(t)|^2 dt \quad (\text{A.5})$$

where A.C. means for absolute continuous. When the potential V is radial, then this coincides with the definition A.1: $\Phi_E^{R_0}(r) = \inf_{y \in S_{R_0}} d_E(x, y)$ for any $x \in \mathbb{R}^d$ with $|x| = r$, where $S_{R_0} := \{y \in \mathbb{R}^d \mid |y| \leq R_0\}$.

We have the following property that will be useful for whose proof we refer to [HS96] proposition 3.3.

Proposition A.3. *For every $R_0 \geq 0$, $r \in [R_0, +\infty) \mapsto \Phi_E^{R_0}$ is differentiable almost everywhere. Where it is differentiable, $|\Phi_E^{R_0'}(r)| \leq (V(r) - E)^+$.*

To get the pointwise bound, the strategy is well known (see e.g. [HS96]) and consits to prove that $e^{\Phi_E^{R_0}} u$ is in $H^1([R, +\infty))$. Indeed, the pointwise bound is obtained by the classical Sobolev embedding $H^1([R, +\infty)) \hookrightarrow C^0([R, +\infty))$. The theorem we are going to prove is therefore

Theorem A.4. *Let u a solution to equation (A.1), η a smooth function which vanishes for $r < R$ and is equal to 1 for $r > R + 1$. We have*

$$\left\| \eta e^{(1-\epsilon)\Phi_E(r)} u \right\|_{H^1} < +\infty. \quad (\text{A.6})$$

Before going into the proof of the bound, let us see what is the Agmon metric for our case of study.

Proposition A.5. *Assume that $\sqrt{\rho} \in \mathcal{L}^{rad}$ with $\int_0^{+\infty} 4\pi r^2 \rho(r) dr = N$ and consider the reduced-Hartree Fock model $V(r) = \frac{l(l+1)}{2r^2} + \frac{-Z}{r} + V^H(\rho)(r)$.*

1. *If $E < 0$,*

$$\Phi_E(r, R_0) = \sqrt{-2E}r + \frac{N-Z}{\sqrt{-2E}} \log(r) + o_{r \rightarrow +\infty}(\log(r)). \quad (\text{A.7})$$

2. If $E = 0$ and $N - Z > 0$,

$$\Phi_E(r, R_0) = 2\sqrt{2(N - Z)r} + o_{r \rightarrow +\infty}(\sqrt{r}). \quad (\text{A.8})$$

Proof. First, we focus on the Hartree potential. Let us recall the formula

$$V^H(\rho)(r) = 4\pi \left(\frac{1}{r} \int_0^r \rho(s)s^2 ds + \int_r^{+\infty} \rho(s)s ds \right).$$

We have then the following inequality

$$|rV^H(\rho)(r) - N| \leq \left| \int_0^r 4\pi s^2 \rho(s) ds - N \right| + \left| \int_r^{+\infty} 4\pi s^2 \rho(s) ds \right|.$$

so that

$$V^H(\rho)(r) \underset{r \rightarrow +\infty}{\sim} \frac{N}{r}.$$

Therefore, we have

$$2(V(r) - E) = \frac{2Q}{r} - 2E + o_{r \rightarrow +\infty}\left(\frac{1}{r}\right)$$

For r big enough, this quantity is positive. If $E < 0$, we have

$$\sqrt{2(V(r) - E)} = \sqrt{-2E} + \frac{Q}{\sqrt{-2E}r} + o_{r \rightarrow +\infty}\left(\frac{1}{r}\right)$$

so that, we get

$$\Phi_E(r, R_0) = \sqrt{-2E}(r - R_0) + \frac{Q}{\sqrt{-2E}} \log\left(\frac{r}{R_0}\right) + o_{r \rightarrow +\infty}(\log(r)).$$

If $E = 0$ and $N - Z > 0$, then

$$\Phi_E(r, R_0) = \sqrt{-2E}(r - R_0) + \frac{Q}{\sqrt{-2E}} \log\left(\frac{r}{R_0}\right) + o_{r \rightarrow +\infty}(\log(r)).$$

□

The first observation begins with the following lemma.

Lemma A.6. *For $f \in W^{1,\infty}(\mathbb{R}^+)$, let us consider the gauge transformed $H_f^V := e^f H^V e^{-f}$. We have*

$$H_f^V - E = H^V - E - \frac{1}{2}|f'|^2 \quad (\text{A.9})$$

in the sense of quadratic form in $L^2(\mathbb{R}^+)$. Moreover, for $0 < \delta < \frac{1}{2}$, $H_f - E$ is coercive on $H_0^1(F_{E,\delta}^f)$ where $F_{E,\delta}^f := \{r \in \mathbb{R}^+ \mid V(r) - E - \frac{1}{2}|f'(r)|^2 \geq \delta\}$, i.e we have

$$\delta \|v\|_{H^1}^2 \leq \langle v | (H_f^V - E) v \rangle \quad (\text{A.10})$$

for every $v \in H_0^1(\mathbb{R}^+)$ supported in $F_{E,\delta}^f$.

Proof. First, note that because $f \in W^{1,\infty}(\mathbb{R}^+)$, then $e^{-f}v \in H_0^1(\mathbb{R}^+)$ for any $v \in H_0^1(\mathbb{R}^+)$. Let $v \in H_0^1(\mathbb{R}^+)$. We have

$$\langle v | H_f^V v \rangle = \langle v | H^V v \rangle - \frac{1}{2} \langle v | |f'| v \rangle + \frac{1}{2} \langle v | f'' v + 2f' v' \rangle. \quad (\text{A.11})$$

Now, note that by integration by part

$$\langle v | f'' v + 2f' v' \rangle = -\langle v | f'' v + 2f' v' \rangle \quad (\text{A.12})$$

so that $\langle v | f'' v + 2f' v' \rangle = 0$ which concludes the proof, because the inequality is straightforward. □

Remark A.7. This coercivity property is very useful to get the bound on $\|u\|_{H^1}$.

This coercivity property is very useful but one have to manage to have $[R, +\infty) \subseteq F_{E,\delta}^f$. Moreover, Clearly, to have the optimal bound, one would like to have f' the biggest possible. To be compatible with the coercivity inequality, the optimal f have to be $f'(r) = \sqrt{2(V(r) - E)_+} + \delta$. This leads to the following notion which traditionnaly used to show such estimate.

Lemma A.8. *Let u a solution to equation (A.1), η a smooth function which vanish for $r < R$ and is equal to 1 for $r > R + 1$ and $f \in W^{1,\infty}(\mathbb{R}^+)$. Let $v = \eta e^f u$ We have*

$$\langle v | (H_f^V - E) v \rangle = \langle ((\eta')^2 + 2\eta\eta'f') e^{2f} u | u \rangle. \quad (\text{A.13})$$

Proof. First, we have

$$\langle v | (H_f^V - E) v \rangle = \langle e^f v | (H^V - E) \eta u \rangle = -\frac{1}{2} \langle e^{2f} \eta u | \eta'' u + 2u' \eta' \rangle \quad (\text{A.14})$$

using $(H^V - E)u = 0$. By integration by part, we compute

$$\langle e^{2f} \eta u | 2u' \eta' \rangle = -2 \langle e^{2f} [(\eta')^2 + \eta\eta'' + 2f'\eta\eta'] u | u \rangle - 2 \langle e^{2f} \eta\eta' u' | u \rangle \quad (\text{A.15})$$

so that combining with equation (A.14), the result follows. $\square$

We can now prove the theorem

Proof. First, to apply the two previous lemma, one need to have $f = (1-\epsilon)\Phi_E(r)$ which is not always the case. We introduce the bounded from abvoe $f_\alpha = \frac{f}{1+\alpha f}$ for $\alpha > 0$. We have $|f_\alpha| \leq \min(\alpha^{-1}, |f|)$ and $|f'_\alpha| \leq |f'|$. Using lemma A.6 and lemma A.8, we have for $v_\alpha = \eta e^{f_\alpha} u$:

$$\delta \|v_\alpha\|_{H^1}^2 \leq \langle v_\alpha | (H_f^V - E) v_\alpha \rangle \quad (\text{A.16})$$

$$\leq \sup_{r \in [R, R+1]} |(\eta'(r) + 2\eta(r)f'_\alpha(r)) \eta'(r) e^{2f_\alpha}| \|u\|^2 \quad (\text{A.17})$$

$\square$

B Understanding the periodic table of elements

Theorem B.1. *Soit $Z = N \in [1, 2]$. $H_{Z,Z}^{0,\text{rHF}}$ has exactly one negative eigenvalue.*

Proof. The number $N(H_{Z,Z}^{0,\text{rHF}}) \in \mathbb{N}$ of negative eigenvalues of $H_{Z,Z}^{0,\text{rHF}}$ is the sum of the number of negative eigenvalues N_l of $H_{Z,Z}^{l,0,\text{rHF}}$ in each section $\mathcal{L}_l$ for $l \geq 0$. The Bargmann inequality gives

$$N_l < \frac{1}{2l+1} \int_0^{+\infty} r |W(r)| dr, \quad (\text{B.1})$$

with $W(r) := -\frac{Z}{r} + V^H(\rho_{Z,Z}^{0,\text{rHF}}(r))$. To shorten the notation, we note ρ instead $\rho_{Z,Z}^{0,\text{rHF}}$ of in the following. Let us recall that $W(r) \leq 0$:

$$\frac{rV^H(\rho)}{4\pi} = \int_0^r s^2 \rho(s) ds + r \int_r^{+\infty} s \rho(s) ds \quad (\text{B.2})$$

$$= Z - \int_r^{+\infty} s(r-s) \rho(s) ds \leq Z \quad (\text{B.3})$$

because $s \geq r$. Now, let us compute

$$\begin{aligned} \int_0^{+\infty} -rW(r) dr &= 4\pi \int_0^{+\infty} \int_r^{+\infty} s(s-r) \rho(s) ds dr \\ &= 4\pi \int_0^{+\infty} \int_0^s s(s-r) dr \rho(s) ds \\ &= 2\pi \int_0^{+\infty} s^3 \rho(s) ds = \frac{1}{2} \int_{\mathbb{R}^3} |x| \rho(x) dx. \end{aligned}$$

We want to show that only the 1s-layer exists. For that, it is enough to show that $\int_{\mathbb{R}^3} |x| \rho(x) dx < 4$. Indeed, this imply $N_0 < 2$ and $N_l < 1$ for every $l \geq 1$. Numerically, we compute $\int_{\mathbb{R}^3} |x| \rho(x) dx \approx 2.5$. $\square$

Energy-dependent Birman–Schwinger principle Let $E = -K^2 < 0$ with $K > 0$. We define the energy-dependent Birman–Schwinger operator

$$B_{\ell,K} = |V_-|^{1/2} (H_{\ell,0} + K^2)^{-1} |V_-|^{1/2}.$$

Theorem B.2 (Energy-dependent Birman–Schwinger principle). *For $K > 0$, the following are equivalent:*

- $-K^2$ is an eigenvalue of H_ℓ ,
- 1 is an eigenvalue of $B_{\ell,K}$.

Moreover, the number of eigenvalues of H_ℓ below $-K^2$ satisfies

$$n_\ell(K) := \#\{\lambda_j(H_\ell) < -K^2\} = \#\{\lambda_j(B_{\ell,K}) > 1\}.$$

Thus $B_{\ell,K}$ counts bound states with energy below $-K^2$. This is particularly useful to exclude near-threshold states.

Integral kernel The resolvent kernel of $(H_{\ell,0} + K^2)^{-1}$ is explicit:

$$G_{\ell,K}(r, s) = \frac{1}{K} \frac{I_{\ell+1/2}(Kr_<) K_{\ell+1/2}(Kr_>)}{\sqrt{rs}},$$

where I_ν and K_ν denote modified Bessel functions and $r_< = \min(r, s)$, $r_> = \max(r, s)$.

Hence

$$(B_{\ell,K} f)(r) = \int_0^\infty V_-(r)^{1/2} G_{\ell,K}(r, s) V_-(s)^{1/2} f(s) ds.$$

Trace bounds As before,

$$n_\ell(K) \leq \text{Tr}(B_{\ell,K}^p), \quad p \geq 1.$$

In particular,

$$n_\ell(K) \leq \text{Tr}(B_{\ell,K}^2) = \iint_0^\infty V_-(r) V_-(s) G_{\ell,K}(r, s)^2 dr ds.$$

By choosing K between two consecutive eigenvalues, one can prove the existence of exactly one bound state in a given energy window.

Zero-energy limit As $K \rightarrow 0^+$, one recovers the zero-energy kernel

$$G_{\ell,K}(r, s) \longrightarrow \frac{1}{2\ell+1} r_<^{\ell+1} r_>^{-\ell},$$

and thus $B_{\ell,K} \rightarrow B_{\ell,0}$, recovering the classical Birman–Schwinger formulation.

Interpretation The energy-dependent formulation allows to:

- count states below a prescribed threshold,
- separate deeply bound states from near-threshold states,
- obtain sharper shell-counting results for large nuclear charge.